In crankshaft metalic knocking sound is appeared?

A **metallic knocking sound from the crankshaft area** usually indicates a problem in the **bottom end of the engine**. It often happens when components connected to the crankshaft develop **excess clearance, wear, or lubrication problems**.

Here are the **most common causes**:

---

1. Worn Connecting Rod Bearings (Rod Knock)

This is the **most common cause** of metallic knocking.

What happens:

- The connecting rod is attached to the crankshaft with a bearing.
- If the bearing wears out, there is **extra clearance between the rod and crankshaft journal**.
- The rod **hits the crankshaft each rotation**, creating a **sharp metallic knock**.

Symptoms

- Deep knocking sound from the engine bottom
- Noise increases with **engine RPM**
- Loud during **acceleration**
- Low oil pressure may appear

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2. Main Bearing Wear

The **crankshaft rotates inside main bearings** in the engine block.

If worn:

- Crankshaft moves slightly
- Produces **heavy metallic knocking or rumbling**

Symptoms

- Low oil pressure
- Constant knocking sound
- Engine vibration

---

3. Crankshaft Journal Damage

If the crankshaft journals become:

- **Scored**
- **Out-of-round**
- **Overheated**

the bearings cannot support the crankshaft properly.

Result

- Metal-to-metal contact
- Knocking noise

---

4. Low Engine Oil / Poor Lubrication

Crankshaft bearings require **hydrodynamic oil film**.

If oil pressure drops:

- Bearings lose lubrication
- Crankshaft contacts metal surfaces
- Knocking begins

Causes

- Low oil level
- Oil pump failure
- Blocked oil passages

---

5. Piston Slap (Sometimes confused with crank knock)

This is from the **piston moving loosely in the cylinder**.

Difference

- Noise is louder during **cold start**
- Comes from **top of engine**, not crankshaft

How to Identify Crankshaft Knock

Typical characteristics:

Feature	Crankshaft/Rod Knock
Sound	Deep metallic knock
RPM effect	Increases with RPM
Location	Bottom of engine
Load effect	Louder during acceleration

Simple Diagnostic Tests

Mechanics usually check:

1. **Oil pressure test**
2. **Oil filter inspection for metal particles**
3. **Engine stethoscope listening**
4. **Cylinder drop test**

If the Crankshaft Is the Problem

Repairs usually require:

- Engine disassembly
- Crankshaft grinding or replacement
- New bearings
- Sometimes **full engine rebuild**

💡 **Important:**
If you keep driving with crankshaft knocking, it can lead to **engine seizure or thrown connecting rod**, which may completely destroy the engine.

If you want, I can also explain:

- **How to distinguish rod knock vs piston knock vs valve train noise**
- **Why VVT engines sometimes create knocking sounds**
- **A diagram of crankshaft bearings and rods**.

Why VVT engines sometimes create knocking sounds

Engines with **Variable Valve Timing (VVT)** sometimes produce **knocking or metallic ticking sounds** because the system uses **oil pressure and mechanical actuators** to change valve timing. If any part of this system fails or lacks proper oil pressure, it can create noise.

Below are the main reasons.

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1. Low Engine Oil Pressure

Most VVT systems operate using **engine oil pressure** to move the camshaft phaser.

What happens

- Low oil level or dirty oil reduces hydraulic pressure.
- The VVT actuator cannot hold its position.
- Internal parts move loosely and create **knocking or rattling**.

Symptoms

- Knocking at startup
- Rough idle

- Engine warning light
- Noise that decreases after oil warms up

---

2. Worn Camshaft Phaser

The **camshaft phaser** adjusts the camshaft timing.

If the internal locking pin or gears wear:

- The camshaft can **oscillate slightly**
- This produces a **metallic rattle or knock**

Typical behavior

- Loud **during idle**
- More noticeable **during acceleration**

---

3. Faulty VVT Solenoid

The **VVT solenoid** controls oil flow to the cam phaser.

If it becomes:

- clogged with sludge
- electrically faulty

then oil flow becomes irregular.

Result

- unstable cam timing
- knocking or ticking noise

---

4. Timing Chain Slack

Many VVT engines use a **timing chain**.

If the chain tensioner or guides wear:

- chain becomes loose
- VVT adjustments make the chain **slap against guides**

This creates a **metallic knocking sound**.

---

5. Oil Sludge in VVT Passages

Small oil passages feed the VVT system.

If oil is not changed regularly:

- sludge blocks passages
- VVT actuator moves incorrectly
- knocking or rattling appears

Typical VVT Noise Characteristics

Condition	Noise Type
Cold start	Short metallic rattle
Low oil pressure	Continuous ticking
Worn cam phaser	Knocking at idle
Timing chain wear	Chain slapping noise

How to Fix VVT Knocking

Common solutions include:

1. Change engine oil and filter
2. Clean or replace VVT solenoid
3. Replace camshaft phaser
4. Replace timing chain and tensioner

✅ **Important:**

VVT knocking is usually **not as deep as crankshaft rod knock**.

- **Rod knock** = deep heavy knock from bottom of engine
- **VVT knock** = lighter metallic ticking from top of engine

---

If you want, I can also show you:

- **A diagram of a VVT system**
- **The sensors used in VVT engines**
- **How to diagnose VVT noise step-by-step (like a mechanic).**

↳ The sensors used in VVT engines

?

Engines with **Variable Valve Timing (VVT)** use several sensors so the **engine control unit (ECU)** can adjust the valve timing accurately. These sensors monitor engine position, load, and operating conditions.

Here are the **main sensors used in VVT systems**:

---

1. Camshaft Position Sensor (CMP)

The **Camshaft Position Sensor** detects the **exact position and rotation speed of the camshaft**.

Purpose

- Tells the ECU the **current valve timing**
- Helps control the VVT actuator

Symptoms if faulty

- Engine misfire

- Rough idle
- VVT malfunction
- Check engine light

---

2. Crankshaft Position Sensor (CKP)

The **Crankshaft Position Sensor** measures the **crankshaft speed and position**.

Purpose

- Synchronizes crankshaft and camshaft
- Allows ECU to calculate correct **valve timing adjustment**

Symptoms if faulty

- Engine may not start
- Irregular timing
- Knocking or poor performance

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3. Mass Air Flow Sensor (MAF)

The **Mass Air Flow Sensor** measures how much air enters the engine.

Purpose

- Determines engine load
- ECU adjusts valve timing for **power or fuel efficiency**

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4. Throttle Position Sensor (TPS)

The **Throttle Position Sensor** monitors the **throttle opening**.

Purpose

- Detects acceleration demand
- ECU advances or retards valve timing for better power

---

5. Engine Oil Pressure Sensor

The **Engine Oil Pressure Sensor** measures oil pressure.

Purpose

- VVT actuators use **hydraulic oil pressure**
- ECU checks if enough pressure exists for proper operation

---

6. Engine Coolant Temperature Sensor

The **Engine Coolant Temperature Sensor** measures engine temperature.

Purpose

- VVT timing changes depending on **engine temperature**
- Helps control emissions and fuel efficiency

✅ Summary of VVT Sensors

Sensor	Function
Camshaft Position Sensor	Detects camshaft angle
Crankshaft Position Sensor	Detects crankshaft rotation
Mass Air Flow Sensor	Measures intake air
Throttle Position Sensor	Detects driver acceleration
Oil Pressure Sensor	Ensures hydraulic pressure
Coolant Temperature Sensor	Monitors engine temperature

💡 Important:

The **two most critical sensors for VVT** are:

- **Camshaft Position Sensor**

- **Crankshaft Position Sensor**

These two sensors allow the ECU to **compare camshaft and crankshaft timing** and control the VVT system precisely.

If you'd like, I can also explain:

- **How the ECU controls the VVT actuator step-by-step**
- **Why VVT sometimes causes metallic knocking at high RPM**
- **A full diagram of a VVT engine system.**

Input side camshaft is structured 2 shaft?

Yes — in many **VVT intake camshaft systems**, the **intake camshaft assembly can effectively have two rotating parts** (not exactly two completely separate shafts, but **two sections that can rotate relative to each other**).

This design allows **Variable Valve Timing to change the valve timing**.

Structure of a VVT Intake Camshaft

A typical intake camshaft with VVT has:

1. **Outer sprocket housing**
2. **Inner camshaft**

These two parts are connected by a **cam phaser mechanism**.

Main Parts

- Timing chain sprocket
- Cam phaser housing
- Inner camshaft
- Lock pin
- Oil chambers

So the camshaft assembly works like **two rotating components**:

Part	Function
Outer part (sprocket housing)	Driven by timing chain
Inner camshaft	Operates intake valves

How It Works

Engine oil pressure is controlled by the **VVT solenoid**.

Oil pressure enters chambers inside the cam phaser and causes **relative rotation between the outer housing and the inner camshaft**.

This changes valve timing.

Example:

- **Advance timing** → intake valve opens earlier
- **Retard timing** → intake valve opens later

Why This Design Is Used

This structure allows the engine to:

- Improve **fuel efficiency**
- Increase **power at high RPM**
- Reduce **emissions**

Why It Sometimes Makes Knocking Noise

Because there are **internal moving parts**, wear can cause:

- cam phaser play
- lock pin failure
- internal gear wear

Then a **metallic knocking or rattling sound** can appear.

✅ Simple idea

```
Timing chain → Outer sprocket
                  ↓
            Cam phaser
                  ↓
            Inner camshaft → opens valves
```

The **outer and inner parts can rotate slightly relative to each other**.

If you want, I can also show you:

- **A clear mechanical diagram of a VVT cam phaser**
- **Why VVT knocking becomes louder at high RPM (important diagnostic clue)**.

I mean, nissan armada 2023 v8 engine has 3 shafts in header

Yes — you are correct. The **2023 Nissan Armada V8 engine** actually has **three shafts in the cylinder head** on each bank. This is because the engine uses both **VVT and variable valve lift technology**.

The engine used is the **5.6-L Nissan VK56VD**, and it includes **Variable Valve Event and Lift (VVEL)**.

Why There Are 3 Shafts in the Cylinder Head

In a normal DOHC engine you usually see **2 camshafts**:

1. Intake camshaft
2. Exhaust camshaft

But in the **VK56VD**, Nissan adds **another control shaft** for the VVEL mechanism.

So the three shafts are:

Shaft	Function
Exhaust camshaft	Opens exhaust valves
Intake camshaft	Drives the VVEL rocker mechanism
VVEL control shaft	Adjusts valve lift continuously

How the 3-Shaft System Works

1. Exhaust Camshaft

- Works like a normal camshaft.
- Opens and closes exhaust valves.

2. Intake Camshaft

- Does **not directly open the intake valves**.
- Instead it drives a **rocker arm system**.

3. VVEL Control Shaft

- This shaft changes the **geometry of the rocker arms**.
- That changes **how much the intake valve opens (valve lift)**.

So the intake valve lift is **continuously adjustable**, not fixed like a normal camshaft.

Simplified Mechanism

```
Intake camshaft
      ↓
Rocker arm linkage
      ↓
VVEL control shaft (adjusts lift)
      ↓
Intake valve
```

The control shaft rotates slightly to **change valve lift from very small to very large**.

Why Nissan Uses This Design

Benefits:

- Better **fuel efficiency**
- Strong **low-RPM torque**
- Higher **high-RPM power**
- Reduced **throttle losses**

Because valve lift is controlled, the engine sometimes **doesn't need a throttle plate as much**.

Why Noise Can Sometimes Appear

If there is wear or oil problems in this system, you may hear:

- **Ticking from rocker arms**
- **Knocking from VVEL linkage**
- **Rattle from cam phaser**

Common causes:

- Low oil pressure
- Worn VVEL actuator
- Loose rocker arm linkage

✅ **So your observation is correct:**
The **Armada V8 cylinder head shows 3 shafts because of the VVEL valve lift system**, not just traditional intake and exhaust cams.

If you want, I can also show you (very interesting for understanding this engine):

- A **diagram of the VK56VD 3-shaft cylinder head**
- Why **Nissan VVEL engines sometimes make ticking noises**
- The **difference between VVEL and normal VVT systems**.

↳ Variable Valve Event and Lift

The purpose?

The purpose of **Variable Valve Event and Lift** in the **Nissan VK56VD** (used in the **Nissan Armada**) is to **continuously control how much and when the intake valves open**.

In simple terms, it allows the engine to **change valve lift and timing depending on driving conditions**.

Main Purposes of VVEL

1. Improve Fuel Efficiency

At **low load or cruising**:

- Intake valves open **a small amount**
- Less air enters the engine
- Fuel consumption decreases

This helps reduce pumping losses that normally occur with a throttle plate.

---

2. Increase Engine Power

At **high RPM or heavy acceleration**:

- Intake valves open **much wider**
- More air enters the cylinders
- More fuel can burn
- Engine produces **more power**

---

3. Better Throttle Response

Because the system controls **airflow directly at the valve**, the engine responds **faster to throttle input**.

Instead of waiting for the throttle plate to open, the engine simply **opens the intake valves more**.

---

4. Reduce Emissions

Precise control of valve opening helps:

- improve air-fuel mixing
- reduce unburned fuel
- lower exhaust emissions

Why the Third Shaft Is Needed

The extra shaft in the cylinder head **changes the rocker arm geometry**, which changes the **valve lift**.

So the three shafts are used to control:

- cam rotation
- rocker movement
- valve lift adjustment

Simple Comparison

System	What It Changes
Traditional camshaft	Fixed valve lift
Variable Valve Timing	Changes valve timing
VVEL	Changes **timing + valve lift continuously**

✅ **In short:**

VVEL is used to **improve power, fuel economy, throttle response, and emissions by precisely controlling intake valve lift and timing.**

If you're interested, I can also explain something very interesting about this engine:
why some Nissan VK56VD engines sometimes produce a ticking or knocking noise from the VVEL mechanism.

What are there electronic or electrical components in engine?

Modern engines (like the V8 in the **Nissan Armada**) contain many **electronic and electrical components**. These components allow the **Engine Control Unit** to monitor the engine and control fuel, ignition, air, and valve timing.

Below are the main categories.

---

1. Sensors (Input Devices)

Sensors **measure engine conditions** and send signals to the ECU.

Position and Speed Sensors

- **Crankshaft Position Sensor** – detects crankshaft rotation and engine speed (RPM).
- **Camshaft Position Sensor** – detects camshaft position for valve timing.

Air Intake Sensors

- **Mass Air Flow Sensor (MAF)** – measures amount of air entering engine.
- **Throttle Position Sensor (TPS)** – measures throttle opening.

Temperature Sensors

- **Engine Coolant Temperature Sensor** – measures engine temperature.

- **Intake Air Temperature Sensor** – measures air temperature entering engine.

Exhaust Sensors

- **Oxygen Sensor** – measures oxygen in exhaust for air-fuel control.
- **Knock Sensor** – detects engine knocking.

Pressure Sensors

- **Manifold Absolute Pressure Sensor (MAP)** – measures intake manifold pressure.
- **Engine Oil Pressure Sensor** – monitors lubrication pressure.

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2. Actuators (Output Devices)

These components **perform actions when commanded by the ECU**.

Fuel System

- **Fuel Injector** – sprays fuel into the engine.

Ignition System

- **Ignition Coil** – generates high voltage for spark plugs.

Air Control

- **Electronic Throttle Body** – controls airflow electronically.

Valve Timing Control

- **VVT Solenoid** – controls oil flow for valve timing adjustment.

Emission Control

- **EGR Valve** – recirculates exhaust gas to reduce emissions.

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3. Control Unit

The **brain of the engine** is the:

- **Engine Control Unit**

It:

- receives signals from sensors

- processes data
- controls actuators

The ECU controls:

- fuel injection timing
- ignition timing
- valve timing
- idle speed
- emissions systems

---

4. Electrical Power Components

These supply electricity to the engine systems.

- **Car Battery** – stores electrical energy.
- **Alternator** – generates electricity while engine runs.
- **Starter Motor** – cranks the engine for starting.

✅ Summary

Category	Examples
Sensors	CKP, CMP, MAF, TPS, O_2, Knock
Actuators	Injectors, ignition coils, throttle body
Control	ECU
Power	Battery, alternator, starter

💡 Since you were asking earlier about **VVT and engine knocking**, the **most important electronic parts related to that** are:

- **Knock Sensor**
- **Camshaft Position Sensor**
- **VVT Solenoid**

If you'd like, I can also explain something very useful for diagnostics:
the 12 most important sensors in modern engines and what happens when each one fails. 🔧🚗

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